GARCH and stochastic volatility models

Christian Francq

Chapter 2: Asymmetric GARCH

Outline

Empirical observation : the volatility tends to increase more after a price drop than after a price increase of the same magnitude.

Symmetry of the standard GARCH

If (ϵ_t) follows a GARCH with a symmetric law for η_t

$$\text{cov}(\sigma_t, \epsilon_{t-h}) = 0, \quad h > 0.$$

Proof

Moreover

$$\text{cov}(\sigma_t, \epsilon_{t-h}) = 0 \quad \Leftrightarrow \quad \text{cov}(\epsilon_t^+, \epsilon_{t-h}) = \text{cov}(\epsilon_t^-, \epsilon_{t-h}) = 0, \quad h > 0,$$

where

$$\epsilon_t^+ = \max(\epsilon_t, 0), \quad \epsilon_t^- = \min(\epsilon_t, 0).$$

Empirical autocorrelations of the CAC 40 (1988-98)

	1	2	3	4	5	10	20
$\hat{\rho}(\epsilon_t, \epsilon_{t-h})$	0.030	0.005	−0.032	0.028	−0.046	0.016	0.003
$\hat{\rho}(\lvert\epsilon_t\rvert, \lvert\epsilon_{t-h}\rvert)$	0.090	0.100	0.118	0.099	0.086	0.118	0.055
$\hat{\rho}(\epsilon_t^+, \epsilon_{t-h})$	0.011	−0.094	−0.148	−0.018	−0.127	−0.039	−0.026

The significant parameters at the level 5% are displayed in red, using $1/n$ as approximation of the variance of the empirical autocorrelations, for $n = 2380$.

Exponential GARCH (EGARCH) of Nelson (1991)

$$\begin{cases} \epsilon_t &= \sigma_t \eta_t \\ \log \sigma_t^2 &= \omega + \sum_{i=1}^{q} \alpha_i g(\eta_{t-i}) + \sum_{j=1}^{p} \beta_j \log \sigma_{t-j}^2 \end{cases}$$

where

$$g(\eta_{t-i}) = \theta \eta_{t-i} + \gamma[|\eta_{t-i}| - E|\eta_{t-i}|].$$

- Multiplicative model for the volatility

$$\sigma_t^2 = e^{\omega} \prod_{i=1}^{q} \exp\{\alpha_i g(\eta_{t-i})\} \prod_{j=1}^{p} \left(\sigma_{t-j}^2\right)^{\beta_j}$$

- No positivity constraint on the coefficients, but

$$-\gamma < \theta < \gamma, \quad \alpha_i \geq 0, \quad \beta_j \geq 0$$

implies that σ_t^2 increases with the modulus of the past innovations (when the sign is fixed).

Exponential GARCH (EGARCH)

$$\begin{cases} \epsilon_t & = \sigma_t \eta_t \\ \log \sigma_t^2 & = \omega + \sum_{i=1}^{q} \alpha_i g(\eta_{t-i}) + \sum_{j=1}^{p} \beta_j \log \sigma_{t-j}^2 \end{cases}$$

where

$$g(\eta_{t-i}) = \theta \eta_{t-i} + \gamma [|\eta_{t-i}| - E|\eta_{t-i}|].$$

- Identifiability problem : $\gamma = 1$.
- Asymmetry taken into account by θ.
  The usual asymmetry observed on the financial series imposes $\theta < 0$.
- The volatility σ_t^2 is expressed as function of the normalized innovations (η_{t-i}).
  $\log \sigma_t^2$ is a strong ARMA since $(g(\eta_t))$ is an iid white noise with variance

$$\mathrm{Var}[g(\eta_t)] = \theta^2 + \gamma^2 \mathrm{Var}(|\eta_t|) + 2\theta\gamma \mathrm{Cov}(\eta_t, |\eta_t|).$$

Difficulties of the statistical inference

- Strict stationarity : if $\beta(z) = 1 - \sum_{i=1}^{p} \beta_i z^i$ has its roots outside the unit circle, then $\sigma_t^2 = \varphi(\eta_u, u < t)$.
- Invertibility : σ_t^2 is a complicated function of $\{\epsilon_u, u < t\}$

$$\log \sigma_t^2 = \omega + \alpha g\left(\frac{\epsilon_{t-1}}{\sigma_{t-1}}\right) + \beta \log \sigma_{t-1}^2.$$

$\rightsquigarrow$ difficult to make estimations and forecasts (cf O. Wintenberger, 2013 `http://fr.arxiv.org/abs/1211.3292`).

Invertibility issue of the EGARCH

Consider the EGARCH(1,1) model

$$\begin{cases} \epsilon_t & = \sigma_t \eta_t \\ \log \sigma_t^2 & = \omega + \theta \eta_{t-1} + \varsigma |\eta_{t-1}| + \beta \log \sigma_{t-1}^2, \end{cases}$$

with η_t iid $\mathcal{N}(0,1)$ and two parameter sets :

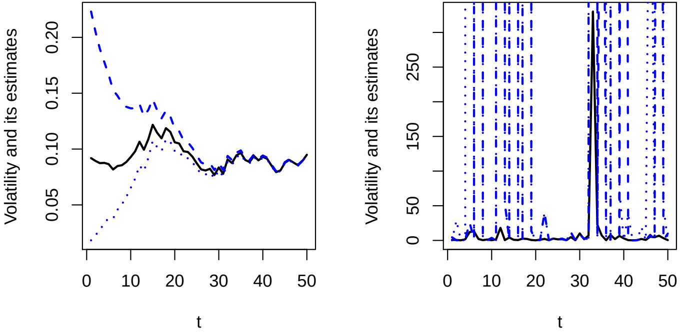
$$\begin{aligned} \text{Design A} \quad &: \quad \omega = -0.3,\ \theta = -0.02,\ \varsigma = 0.20,\ \beta = 0.97 \\ \text{Design B} \quad &: \quad \omega = -4,\ \theta = -0.02,\ \varsigma = 5.8,\ \beta = 0.2. \end{aligned}$$

The two models are stationary because $|\beta| < 1$.
For some initial value $h \in \mathbb{R}$, let $\hat{h}_1(h) = h$ and

$$\hat{h}_t(h) = \log \hat{\sigma}_t^2(h) = \omega + (\theta \epsilon_{t-1} + \varsigma |\epsilon_{t-1}|) e^{-\frac{\hat{h}_{t-1}(h)}{2}} + \beta \hat{h}_{t-1}(h),$$

for $t = 2, 3, \ldots$

In Design B, the EGARCH is not invertible



Volatility σ_t^2 (in full line) and volatility estimates $\hat{\sigma}_t^2(h)$ (in dashed and dotted lines) for two different initial values h, when the simulated EGARCH model is invertible (Design A, left panel) or non invertible (Design B, right panel).

$$\frac{\sum_{i=1}^{q}\alpha_i L^i}{\beta(L)} = \sum_{i=1}^{\infty}\lambda_i L^i, \qquad g_\eta(x) = E[\exp\{xg(\eta_t)\}]$$

Gaussian case

Moments of the EGARCH(p,q)

Let m be a positive integer. Under the strict stationarity conditions and if

$$\mu_{2m} = E(\eta_t^{2m}) < \infty, \qquad \prod_{i=1}^{\infty} g_\eta(m\lambda_i) < \infty,$$

(ϵ_t^2) admits a moment of order m

$$E(\epsilon_t^{2m}) = \mu_{2m} e^{m\omega^*} \prod_{i=1}^{\infty} g_\eta(m\lambda_i).$$

In the Gaussian case, all the moments exist, which is not in adequacy with the leptokurticity of the financial series.

Threshold GARCH (TGARCH or ZARCH of Zakoïan, 1994)

$$\begin{cases} \epsilon_t & = \sigma_t \eta_t \\ \sigma_t & = \omega + \sum_{i=1}^{q} \alpha_{i,+}\epsilon_{t-i}^{+} - \alpha_{i,-}\epsilon_{t-i}^{-} + \sum_{j=1}^{p} \beta_j \sigma_{t-j} \end{cases}$$

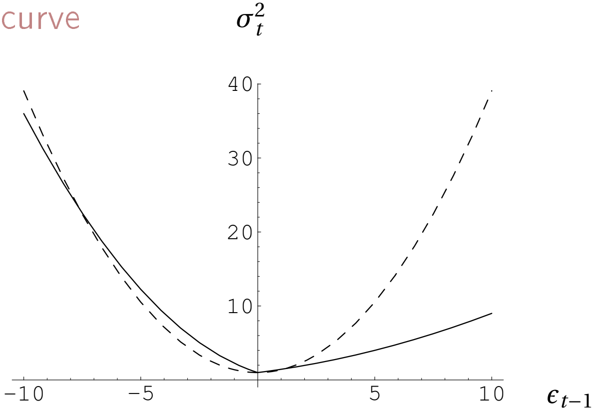
- Positivity constraints

$$\omega > 0, \quad \alpha_{i,+} \geq 0, \quad \alpha_{i,-} \geq 0, \quad \beta_i \geq 0.$$

- Through $\alpha_{i,+}$ and $\alpha_{i,-}$, the volatility depends on the sign of the past innovations.
- Symmetric model if for all $i = 1, \ldots, q$, $\alpha_{i,+} = \alpha_{i,-} = \alpha_i$:

$$\sigma_t = \omega + \sum_{i=1}^{q} \alpha_i |\epsilon_{t-i}| + \sum_{j=1}^{p} \beta_j \sigma_{t-j}$$

News impact curve



News impact curves of the ARCH(1) : $\epsilon_t = \sqrt{1+0.38\epsilon_{t-1}^2}\eta_t$ (dotted line) and TARCH(1) : $\epsilon_t = (1-0.5\epsilon_{t-1}^- + 0.2\epsilon_{t-1}^+)\eta_t$ (full line).

Stationnarity

Noting that

$$\epsilon_t^+ = \sigma_t \eta_t^+, \qquad \epsilon_t^- = \sigma_t \eta_t^-$$

we have

$$\sigma_t = \omega + \sum_{i=1}^{\max\{p,q\}} a_i(\eta_{t-i})\sigma_{t-i}, \qquad a_i(z) = \alpha_{i,+} z^+ - \alpha_{i,-} z^- + \beta_i.$$

When $p = q = 1$ the model admits a unique strictly stationary and nonanticipative solutions iff

$$E\{\log a(\eta_t)\} < 0.$$

TARCH(1) with (η_t) of symmetric law : $\alpha_{1,+}\alpha_{1,-} < e^{-2E\log|\eta_t|}$. *

*. Indeed, $\log \alpha_+ \eta_t^+ - \alpha_- \eta_t^- = 1_{\eta_t > 0} \log \alpha_+ + 1_{\eta_t < 0} \log \alpha_- + \log|\eta_t|$

The TARCH(1,1) $\sigma_t = \omega + \alpha_+ \epsilon_{t-1}^+ - \alpha_- \epsilon_{t-1}^- + \beta \sigma_{t-1}$

admits a second-order stationary solution iff

$$E[(\alpha_+ \eta_t^+ - \alpha_- \eta_t^- + \beta)^2] < 1.^\dagger$$

Asymmetry : assuming in addition that the law of η_t is symmetric, we have

$$\begin{aligned} \operatorname{cov}(\sigma_t, \epsilon_{t-1}) &= E(\omega + \alpha_+ \epsilon_{t-1}^+ - \alpha_- \epsilon_{t-1}^-)\epsilon_{t-1} \\ &= \alpha_+ E(\epsilon_{t-1}^+)^2 - \alpha_- E(\epsilon_{t-1}^-)^2 \\ &= (\alpha_+ - \alpha_-) E(\epsilon_t^+)^2 \neq 0 \end{aligned}$$

when $\alpha_+ \neq \alpha_-$.

---

†. Indeed, if there exists a stationary causal solution to $\sigma_t = \omega + a_{t-1}\sigma_{t-1}$ such that $E\sigma_t^2 < \infty$, then $(1 - Ea_t^2)E\sigma_t^2 = \omega^2 + 2\omega Ea_t E\sigma_t > 0$, which entails $Ea_t^2 < 1$. Conversely, if $Ea_t^2 < 1$, then $E\log a_t < 0$ and the stationary solution satisfies $\|\sigma_t\|_2 \leq \omega(1 + \|a_t\|_2 + \|a_t\|_2^2 + \cdots) < \infty$.

Stationarity of the TGARCH(p,q)

As for the standard GARCH(p,q) we use the representation

$$\underline{z}_t = \underline{b}_t + A_t \underline{z}_{t-1}$$

where

$$\underline{b}_t = \underline{b}(\eta_t) = \begin{pmatrix} \omega\eta_t^+ \\ -\omega\eta_t^- \\ 0 \\ \vdots \\ \omega \\ 0 \\ \vdots \\ 0 \end{pmatrix} \in \mathbb{R}^{p+2q}, \quad \underline{z}_t = \begin{pmatrix} \epsilon_t^+ \\ -\epsilon_t^- \\ \vdots \\ \epsilon_{t-q+1}^+ \\ -\epsilon_{t-q+1}^- \\ \sigma_t \\ \vdots \\ \sigma_{t-p+1} \end{pmatrix} \in \mathbb{R}^{p+2q},$$

Stationarity of the TGARCH(p,q)

$$\underline{z}_t = \underline{b}_t + A_t \underline{z}_{t-1}$$

where

$$A_t = \begin{pmatrix} \alpha_{1,+}\eta_t^+ & \cdots & \alpha_{q,+}\eta_t^+ & \alpha_{q,-}\eta_t^+ & \beta_1\eta_t^+ & \cdots & \beta_p\eta_t^+ \\ -\alpha_{1,+}\eta_t^- & \cdots & -\alpha_{q,+}\eta_t^+ & -\alpha_{q,-}\eta_t^- & -\beta_1\eta_t^- & \cdots & -\beta_p\eta_t^- \\ & I_{2(q-1)} & 0_{2(q-1)\times 1} & 0_{2(q-1)\times 1} & & 0_{2(q-1)\times(p-1)} & 0_{2(q-1)\times 1} \\ \alpha_{1,+} & \cdots & \alpha_{q,+} & \alpha_{q,-} & \beta_1 & \cdots & \beta_p \\ & 0_{(p-1)\times 2(q-1)} & 0_{(p-1)\times 1} & 0_{(p-1)\times 1} & & I_{p-1} & 0_{p-1\times 1} \end{pmatrix}.$$

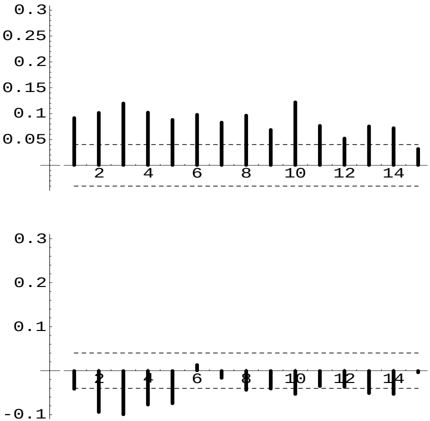
Strict stationarity of the TGARCH(p,q)

As for the standard GARCH(p,q) : strict stationarity iff $\gamma(\mathbf{A}) < 0$.

$$\gamma(\mathbf{A}) = \lim_{t\to\infty} \frac{1}{t} \log \| A_t A_{t-1} \cdots A_1 \|.$$

Estimation and comparison of models for the CAC index

Correlogram $h \mapsto \hat{\rho}(|r_t|, |r_{t-h}|)$ of the absolute CAC 40 returns and cross correlogram $h \mapsto \hat{\rho}(|r_t|, r_{t-h})$ measuring the leverage effect



Fitting different GARCH models

- Standard GARCH(1,1)

$$\begin{cases} r_t = \underset{(2.10^{-4})}{5.10^{-4}} + \epsilon_t, \quad \epsilon_t = \sigma_t \eta_t, \quad \eta_t \sim \mathcal{N}(0,1) \\ \sigma_t^2 = \underset{(2.10^{-6})}{8.10^{-6}} + \underset{(0.02)}{0.09} \epsilon_{t-1}^2 + \underset{(0.02)}{0.84} \sigma_{t-1}^2 \end{cases}$$

- EGARCH(1,1)

$$\begin{cases} r_t = \underset{(2.10^{-4})}{4.10^{-4}} + \epsilon_t, \quad \epsilon_t = \sigma_t \eta_t, \quad \eta_t \sim \mathcal{N}(0,1) \\ \log \sigma_t^2 = \underset{(0.15)}{-0.64} + \underset{(0.03)}{0.15} \left(\underset{(0.14)}{-0.53} \eta_{t-1} + |\eta_{t-1}| - \sqrt{\frac{2}{\pi}} \right) \\ \qquad\qquad + \underset{(0.02)}{0.93} \log \sigma_{t-1}^2 \end{cases}$$

- QGARCH(1,1)

$$\begin{cases} r_t = \underset{(2.10^{-4})}{3.10^{-4}} + \epsilon_t, \quad \epsilon_t = \sigma_t \eta_t, \quad \eta_t \sim \mathcal{N}(0,1) \\ \sigma_t^2 = \underset{(2.10^{-6})}{9.10^{-6}} + \underset{(0.01)}{0.07} \epsilon_{t-1}^2 - \underset{(2.10^{-4})}{9.10^{-4}} \epsilon_{t-1} + \underset{(0.03)}{0.85} \sigma_{t-1}^2 \end{cases}$$

Fitting different GARCH models

- GJR-GARCH(1,1)

$$\begin{cases} r_t = \underset{(2.10^{-4})}{4.10^{-4}} + \epsilon_t, \quad \epsilon_t = \sigma_t \eta_t, \quad \eta_t \sim \mathcal{N}(0,1) \\ \sigma_t^2 = \underset{(2.10^{-6})}{1.10^{-5}} + \underset{(0.02)}{0.13} \epsilon_{t-1}^2 - \underset{(0.02)}{0.10} \epsilon_{t-1}^2 \mathbb{1}_{\{\epsilon_{t-1}>0\}} + \underset{(0.03)}{0.84} \sigma_{t-1}^2 \end{cases}$$

- TGARCH(1,1)

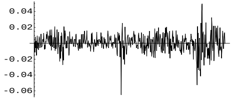
$$\begin{cases} r_t = \underset{(2.10^{-4})}{4.10^{-4}} + \epsilon_t, \quad \epsilon_t = \sigma_t \eta_t, \quad \eta_t \sim \mathcal{N}(0,1) \\ \sigma_t = \underset{(2.10^{-4})}{8.10^{-4}} + \underset{(0.01)}{0.03} \epsilon_{t-1}^+ - \underset{(0.02)}{0.12} \epsilon_{t-1}^- + \underset{(0.02).}{0.87} \sigma_{t-1} \end{cases}$$

Log-likelihood of the different models

	GARCH	EGARCH	QGARCH	GJR-GARCH	TGARCH
$\log L_n$	7393	7404	7404	7406	7405

Volatilities of the estimated models

500 first values of the CAC40 index and estimated volatility ($\times 10^4$) by the GARCH(1,1), EGARCH(1,1), QGARCH(1,1), GJR-GARCH(1,1) and TGARCH(1,1) models

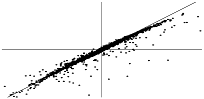


Distance between the estimated models

Mean of the squared differences between the estimated volatilities ($\times 10^{10}$).

	GARCH	EGARCH	QGARCH	GJR	TGARCH
GARCH	0	10.98	3.58	7.64	12.71
EGARCH	10.98	0	3.64	6.47	1.05
QGARCH	3.58	3.64	0	3.25	4.69
GJR	7.64	6.47	3.25	0	9.03
TGARCH	12.71	1.05	4.69	9.03	0

Scatterplot of the estimated volatilities of the EGARCH and TARCH, or TGARCH and GJR-GARCH



Marginal law : real series vs estimated models

Variance ($\times 10^4$) and Kurtosis of the CAC 40 series and of simulations of length 50000 of the 5 estimated models with Gaussian errors.

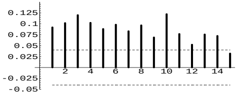
	CAC 40	GARCH	EGARCH	QGARCH	GJR	TGARCH
Kurtosis	5.9	3.5	3.4	3.3	3.6	3.4
Variance	1.3	1.3	1.3	1.3	1.3	1.3

Number of CAC returns ourside $\bar{r} \pm 3\hat{\sigma}_t$
THEO is the theoretical mean value when the conditional distribution is $\mathcal{N}(0, \hat{\sigma}_t^2)$.

THEO	GARCH	EGARCH	QGARCH	GJR	TGARCH
6	17	13	14	15	13

Asymmetries : real series vs estimated models

Correlogram $h \mapsto \rho(|r_t|,|r_{t-h}|)$ of the absolute values (left graphs) and cross correlogram $h \mapsto \rho(|r_t|, r_{t-h})$ (right graphs) : for the CAC 40 index (top), for the standard GARCH (middle) and for the TGARCH (bottom) estimated on the CAC 40 series



End of Chapter 2

Symmetry of the standard GARCH

Note that σ_t is a function of $\eta^2_{t-1},\dots,\eta^2_{t-h}$ and of $\epsilon^2_{t-h-1},\epsilon^2_{t-h-2},\dots$, that is denoted by

$$\mathfrak{h}(\eta^2_{t-1},\dots,\eta^2_{t-h},\epsilon^2_{t-h-1},\epsilon^2_{t-h-2},\dots).$$

Since η_{t-h} is independent of the other variables of $\mathfrak{h}$, we have for all h

$$\begin{aligned} & \mathrm{Cov}(\sigma_t,\epsilon_{t-h}) \\ = \; & E\{E(\sigma_t\epsilon_{t-h}\mid \eta_{t-1},\dots,\eta_{t-h+1},\epsilon_{t-h-1},\epsilon_{t-h-2},\dots)\} \\ = \; & E\left\{\int \mathfrak{h}(\eta^2_{t-1},\dots,\eta^2_{t-h+1},x^2,\epsilon^2_{t-h-1},\epsilon^2_{t-h-2},\dots)x\sigma_{t-h}dP_\eta(x)\right\}=0 \end{aligned}$$

when the distribution of P_η is symmetric.

Return

Computation of $g_\eta(\lambda) = E\exp\{\lambda g(\eta)\}$, $g(\eta) = \theta\eta + \gamma(|\eta| - E|\eta|)$

Let ϕ be the density of $\eta \sim \mathcal{N}(0,1)$,

$$Ee^{\lambda\eta}1_{\{\eta>0\}} = \int_0^\infty e^{\lambda x}\phi(x)\,dx = \int_0^\infty e^{\frac{\lambda^2}{2}}\phi(x-\lambda)\,dx = e^{\frac{\lambda^2}{2}}\Phi(\lambda)$$

and

$$E|\eta| = 2\int_0^\infty x\phi(x)\,dx = \sqrt{2/\pi}.$$

We thus have

$$g_\eta(\lambda) = e^{-\lambda\gamma\sqrt{2/\pi}}\left[e^{\lambda^2(\theta+\gamma)^2/2}\Phi\{\lambda(\theta+\gamma)\} + e^{\lambda^2(\theta-\gamma)^2/2}\Phi\{\lambda(\gamma-\theta)\}\right].$$

Since $\Phi(x) \sim 1/2 + x\phi(0)$ and $\log 2\Phi(x) \sim 2x\phi(0)$ when $x \sim 0$,

$$\log g_\eta(\lambda) = O(\lambda), \quad \lambda \sim 0.$$

Return